

# Developer Salary Prediction

## ML1 Task 2 — Regression

Łukasz Madej    Osman Aliyev

May 2026

# Problem

**Goal:** Predict annual developer compensation (`annual.pay.usd`)

**Metric:** Root Mean Square Error (RMSE)

**Constraint:** ML1-syllabus models only (no trees)

---

|               |                                               |
|---------------|-----------------------------------------------|
| Training rows | ~2 500                                        |
| Test rows     | 628                                           |
| Target        | Annual salary in USD (heavy-tailed)           |
| Key challenge | Multi-select survey columns, extreme outliers |

---

## Three decisions made before any model:

1. Multi-select columns (tech stack, tools, languages) — multi-hot + count features
2. Target modelled in **log-space** to stabilise variance
3. Salary outliers handled by **range filter** [1k, 300k]

# Feature Engineering

**Multi-select columns** (technologies, tools, languages, frameworks):

- ▶ Multi-hot encode each column (`min_freq = 100`, `top_k = 10` per column)
- ▶ Add **count feature** per column — breadth-of-experience proxy
- ▶ Derived: `professional_experience_share`, `non_professional_years`

**Target encoding** (8 high-cardinality columns):

- ▶ `region`, `industry`, `company.size`, `age.group`, `dev.role`, `education`, `employment.type`, `work.location`
- ▶ Smoothed K-fold target encoding — no leakage across folds

**Preprocessing:**

- ▶ `StandardScaler` + mean imputation
- ▶ Salary filter [`1 000`, `300 000`] applied only to training data
- ▶ All steps fitted on training fold only

## Algorithms Considered

5-fold cross-validation on the same pipeline, RMSE in USD:

| Model                                  | Unfiltered data | After range filter |
|----------------------------------------|-----------------|--------------------|
| Ridge                                  | 93 363          | 39 482             |
| Lasso                                  | 90 980          | 38 232             |
| Elastic Net                            | 91 848          | 38 702             |
| KNN                                    | 87 898          | 38 826             |
| <b>SVR (RBF)</b>                       | <b>86 295</b>   | <b>36 659</b>      |
| Random Forest ( <i>reference</i> )     | 84 277          | 34 383             |
| Gradient Boosting ( <i>reference</i> ) | 84 194          | 34 658             |

- ▶ Range filtering alone cuts RMSE by ~55 k — extreme salaries dominate squared error
- ▶ **SVR is the best ML1-allowed model**; tree methods (out of scope) are only ~2 k better
- ▶ Linear models lag significantly — salary signal is non-linear

# Hyperparameter Tuning

Three rounds of tuning for SVR:

1. **Preprocessing sweep** — scaler, imputer, min\_freq / top\_k for multi-hot encoding
2. **SVR grid search** — C in {1, 2, 3, 5}, epsilon in {0.03, 0.05, 0.08}, gamma in {scale, auto, 0.005, 0.01}
3. **Target encoding ablation** — progressively adding high-cardinality columns

| Tuning stage                 | 5-fold CV RMSE |
|------------------------------|----------------|
| Initial SVR (default params) | 36 659         |
| After preprocessing tuning   | 33 395         |
| + SVR grid search            | 31 488         |
| + Target encoding (8 cols)   | <b>31 284</b>  |

**Selected:** SVR(kernel=rbf, C=1, eps=0.03, gamma=auto) with StandardScaler and mean imputation

# Ensembling

## Out-of-fold (OOF) blend of SVR + Random Forest + Ridge:

- ▶ Each model produces OOF predictions on the training set
- ▶ Weights optimised with Nelder-Mead on OOF RMSE
- ▶ Best found blend: **SVR 90.3% + RF 9.7%** (Ridge weight collapsed to  $\sim 0$ )

| Configuration                  | OOF RMSE (eval on [1k, 300k]) |
|--------------------------------|-------------------------------|
| SVR alone                      | 31 284                        |
| SVR + RF + Ridge (Nelder-Mead) | <b>30 984</b>                 |

## Training on full data:

- ▶ Blend weights fixed from OOF optimisation
- ▶ Both SVR and RF retrained on all  $\sim 2\,300$  filtered rows
- ▶ Test predictions: 90.3% SVR + 9.7% RF (on log-scale, then exponentiated)

## Why the Blend

| Argument                               | Detail                                                                                           |
|----------------------------------------|--------------------------------------------------------------------------------------------------|
| OOF gain is real<br>RF complements SVR | -300 RMSE on 5 folds, consistent across splits<br>RF catches mid-range clusters SVR smooths over |
| Tiny RF weight                         | 9.7% limits risk of overfitting RF to training noise                                             |
| Ridge dropped                          | Ridge too weak (38k) to add value alongside SVR (31k)                                            |

### Risks acknowledged:

- ▶ RF is outside the ML1 syllabus — used only as a blend component in the final submission
- ▶ OOF evaluation uses filtered salaries [1k, 300k]; true test set may include extreme values
- ▶ Expected ~2–3 k CV-to-LB gap: range filter before CV split makes CV optimistic

**Bottom line:** SVR alone at ~31 k is the “safe” submission; the blend at ~31 k is a ~300-point improvement with limited downside.

# Expected Results

## Final pipeline (end-to-end):

1. Salary filter [1k, 300k] on training data only
2. Multi-hot encode + count / derived features
3. K-fold target encode 8 high-cardinality columns
4. StandardScaler + mean imputation
5. **SVR** (RBF, C=1, eps=0.03, gamma=auto) fit on log-salary
6. Optional: blend 90.3% SVR + 9.7% RF predictions
7. `exp(prediction)` → submission

| Estimate                                  | Value                                      |
|-------------------------------------------|--------------------------------------------|
| 5-fold CV RMSE (filtered eval, SVR alone) | 31 284                                     |
| 5-fold CV RMSE (OOF blend)                | 30 984                                     |
| Expected leaderboard RMSE                 | ~33–34 k                                   |
| CV-to-LB gap                              | ~2–3 k (structural, range filter artefact) |



## Summary

---

|                    |                                                             |
|--------------------|-------------------------------------------------------------|
| <b>Best model</b>  | SVR with RBF kernel                                         |
| <b>Why SVR</b>     | Best ML1-allowed model; non-linear; robust to feature scale |
| <b>Biggest win</b> | Range filtering — removed extreme salaries (-55 k RMSE)     |
| <b>Second win</b>  | Smoothed K-fold target encoding (-5 k RMSE)                 |
| <b>Blend gain</b>  | SVR 90.3% + RF 9.7% (-300 RMSE over SVR alone)              |
| <b>Expected LB</b> | ~33–34 k RMSE                                               |

---

### Tried and rejected:

- ▶ Winsorization — hurt CV by ~4 k (clipped targets in validation folds)
- ▶ SVR + Ridge blend — Ridge too weak; blend pulled SVR down
- ▶ Stacking (meta-learner) — did not outperform the tuned single SVR

Near the data-driven ceiling for ML1-syllabus models on this dataset.